FEDERAL SPECIFICATION

GRINDING AND POLISHING MACHINE, DENTAL LABORATORY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers machines suitable for the grinding and polishing operations necessary in a dental prosthetic laboratory.

1.2 Classification.—Machines shall be of the following types, as specified (see 3.4.3.1 and 6.2).

Type I.—Capacitor-start.
Type II.—Repulsion-start induction.
Type III.—Split-phase.
Type IV.—Permanent-split capacitor.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND OTHER PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CC-M-636 — Motors, Fractional Horsepower (Alternating-Current).
NN-B-621 — Boxes, Wood, Nailed and Lock-Corner.
PPP-B-601—Boxes, Wood, Cleated-Plywood.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other products specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

JAN-P-106 — Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN-P-125 — Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
MIL-F-13924—Finish, Black Oxide, for Iron and Steel.
MIL-C-16173 — Corrosion Preventive, Solvent Cutback, Cold-Application.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 — Marking for Shipment and Storage.

Armed Services Medical Procurement Agency Drawings:

1056 Insignia.

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2039 Adapter, Chuck, Dental Grinding and Polishing Machine, Right, Automatic.
2040 Chuck, Grinding and Polishing Machine, Dental Laboratory, Arbor Band, Right.
2041 Chuck, Grinding and Polishing Machine, Dental Laboratory, Bur, Right.
2042 Chuck, Grinding and Polishing Machine, Dental Laboratory, Right.
2043 Chuck, Grinding and Polishing Machine, Dental Laboratory, Stone, Right.

(Reduced copies of the drawings are included herein as figures 1, 2, 3, 4, 5, 6. Copies of original size drawing will be furnished manufacturers, upon request, by the procuring agency or as directed by that agency.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publications form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply:

National Board of Fire Underwriters:

NBFU Pamphlet No. 70.—National Electric Code, as amended-Standard of the National Board of Fire Underwriters for Electric Wiring and Apparatus.

(NBFU standards are published by the National Board of Fire Underwriters, 85 John Street, New York 7, N. Y., 222 West Adams Street, Chicago 6, Ill., and Merchants Exchange Building, San Francisco 4, Calif.)

Underwriters' Laboratories, Inc.

Underwriters' Laboratories Standards.

(Underwriters' Laboratories Standards are available from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, N. Y., 207 East Ohio Street, Chicago 11, Ill., and 1655 Scott Lane, Santa Clara, Calif.)

3. REQUIREMENTS

3.1 Conformance with Underwriters' Laboratories Standards.

3.1.1 The contractor shall submit to the procuring agency proof that the equipment he proposes to supply under this specification conforms to the applicable standards of the Underwriters' Laboratories, Inc. Equipment bearing the label or listing of the Underwriters' Laboratories, Inc., will be accepted as evidence of compliance with this requirement.

3.1.2 In lieu of evidence of the Underwriters' Laboratories, Inc., the contractor may submit proof from other sources, acceptable to the purchasing agency, that his commodity conforms to the applicable published standards of the Underwriters' Laboratories, Inc.

3.1.3 Compliance with fire-and-casualty-hazard requirements of Underwriters' Laboratories, Inc., does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity. In the event of difference between any of the requirements of this specification and those of Underwriters' Laboratories, Inc., the requirements of this specification will govern.

3.2 Design and construction.—Unless otherwise specified, grinding and polishing machines shall be designed and constructed for operation on 110-volt, 60-cycle, single-phase, alternating-current.

3.3 Materials.—Materials used in construction of the machines shall be as hereinafter specified. When materials are used which are not specifically designated, they shall be of high commercial quality throughout, and suitable in all respects for the purpose intended. The materials in all cases shall be consistent with the strength required for safety and reliability. Machines in their entirety shall be of such design and material as to assure rigid, rugged, and durable construction, and shall perform in a satisfactory manner the functions for which such machines are intended.

3.4 Components.—Grinding and polishing machines shall consist essentially of housing

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and base, motor frame and end plates, motor, and accessories, as hereinafter specified.

3.4.1 *Housing and base.*—Motor and switch shall be completely enclosed in a smoothly finished cast iron housing that will render it dustproof and dirtproof, and facilitate easy cleaning. The motor shall be supported by a base, cast integrally, or separately fastened to the motor frame. The base shall be of substantial metal with sufficient area to furnish firm support for the motor. Base shall have four supporting feet, each shod with heavy rubber in order to prevent slippage on, or damage to, the bench top.

3.4.2 *Motor frame and end plates.*—Motor frame shall be of smoothly finished metal of substantial thickness. End plates shall be of such type and construction that they may be readily and securely fastened into position without affecting the bearing alignment.

3.4.3 *Motor.*

3.4.3.1 *Type.*—The motor shall conform to the applicable requirements of Federal Specification CC–M–636 for either the capacitor-start type, repulsion-start induction type, split-phase type, or of the permanent-split capacitor type, as specified in the invitation for bids.

3.4.3.2 *Variation from rated voltage.*—Motor shall operate successfully at rated load with a variation in the voltage of plus or minus 10 percent of rated voltage.

3.4.3.3 *Power.*—Motor shall develop not less than ⅙-horsepower at high speed.

3.4.3.4 *Speed.*—Motor shall have not less than two operating speeds. Low speed shall be 1700 to 1750 revolutions per minute (r.p.m.) and high speed shall be 3400 to 3500 r.p.m. The "no load speed" shall not exceed 3600 r.p.m. The speed shall be determined by either change of poles or a mechanically operated governor.

3.4.3.5 *Control switch.*—Control switch shall be of the sliding lever type. Snap action shall be provided at the contacts. The switch shall be located at front of motor or in the base, and shall be totally enclosed, except for slotted opening that may be required by lever action design. If a slot is required, it shall be so designed and located that dust cannot enter housing. A cover plate shall be provided with "off position" and "speed positions" indicated thereon.

3.4.3.6 *Rotor.*—The rotor shall be accurately balanced, statically and dynamically, to provide smooth quiet operation at each speed.

3.4.3.7 *Shaft.*—The shaft shall be hardened steel alloy, and shall extend at each end of the motor for a sufficient length to accommodate standard lathe chucks. The shaft shall be accurately ground to precision limits, and shall be provided with tapered ends for the reception of chucks. The taper shall be ¼ inch per foot with a diameter of approximately 0.355-inch at extreme ends. The shaft shall not move parallel to its long axis, except when chucks are being removed. The center of shaft shall be not less than 4 inches and not more than 5 inches from the surface of workbench.

3.4.3.8 *Bearings.*—Motors may be furnished with either sleeve or ball bearings of suitable design, size, and quality. The bearings shall have permanent lubrication sealed in so that no further lubrication is required for the life of the bearings. The alloy used in construction of sleeve bearings shall have a composition of 70 percent copper, 20 percent lead, and 10 percent tin. Sleeve bearings shall be accurately fitted to the bearing brackets where each shall be located and secured by a tapered-end dowel screw, and shall be removable from the bearing brackets, without tools, upon removal of the tapered dowel screws. Ball bearings shall be accurately fitted into the motor and shall be readily and easily removable. Bearings shall be standardized to permit replacement without machining or fitting. Washers shall be provided at the outer end of each bearing,

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and shall be so arranged as to exclude foreign matter such as pumice, etc., from entering the bearings.

3.4.3.9 *Chuck removing device.* — A suitable chuck removing device shall be located at each end of the shaft to provide easy removal of chuck by turning of lever handle. If machine has ball bearings, the chuck removing device must be designed so that its operation does not exert excess thrust (or chuck load) between the balls and races of the bearing.

3.4.3.10 *Connecting cord.* — Connecting cord shall consist of a rubber-covered service cable with an indestructible type rubber connection plug, and shall be not less than 8 feet in length. The current rating of the cord shall be not less than that of the motor. The cord shall enter rear of motor housing or base through an opening equipped with an insulating bushing.

3.4.3.11 *Noise.*—Motors shall operate with a minimum of noise at all loads and speeds within the service range.

3.4.3.12 *Motor stall requirements.*—Each machine, when connected to a circuit of the rated voltage, frequency, and type of current of the motor, shall show no damage or smoke when the armature or rotor is stalled for a period of 5 seconds.

3.4.4 *Accessories.*—Unless otherwise specified, machines shall be furnished with one automatic adapter, four interchangeable adapter-type chucks for use with the automatic adapter, and two standard chucks, as hereinafter specified.

3.4.4.1 *Adapter, chuck, right, automatic.*—Adapter, chuck, right, automatic shall be for use on the right-hand side of the machine, and shall permit the interchangeability of adapter-type dental chucks while the machine is in operation. The adapter shall have three Allen screws for attachment to the machine shaft and for adjustment for true-running. Each adapter shall be furnished with four extra springs. The dimensions, style, and design of the adapter shall conform to Drawing 2039. (See figure 2). Adapter shall be fabricated of material as specified in this drawing. The finish shall be a black oxide conforming to class 1, grade A of Military Specification MIL–F–13924.

3.4.4.2 *Chuck, arbor band, right, automatic adapter type.* — Chuck, arbor band, right, automatic adapter type shall be a carbon steel adapter-type chuck for emery arbor band, and shall be suitable for use with the automatic adapter as specified in 3.4.4.1. The dimensions, style, and design of the chuck shall conform to Drawing 2040. (See figure 3).

3.4.4.3 *Chuck, bur, right, automatic adapter type.*—Chuck, bur, right, automatic adapter type shall be a carbon steel adapter-type chuck for burs, mandrels, and mandrel mounted silicon carbide points and wheels, and shall be suitable for use with the automatic adapter as specified in 3.4.4.1. The dimensions, style, and design of the chuck shall conform to Drawing 2041. (See figure 4). Chuck shall be furnished with four extra springs.

3.4.4.4 *Chuck, spindle, right, automatic adapter type.*—Chuck, spindle, right, automatic adapter type shall be a carbon steel adapter-type spindle chuck for cloth, felt, and brush wheels, and shall be suitable for use with the automatic adapter as specified in 3.4.4.1. The dimensions, style, and design of the chuck shall conform to Drawing 2042. (See figure 5).

3.4.4.5 *Chuck, stone, right, automatic adapter type.* — Chuck, stone, right, automatic adapter type shall be a carbon steel adapter-type chuck for unmounted stone or rubber lathe wheels, and shall be suitable for use with the automatic adapter as specified in 3.4.4.1. The dimension, style, and design shall conform to Drawing 2043. (See figure 6).

3.4.4.6 *Chuck, stone, left.*—Chuck, stone, left, shall be a standard brass or bronze

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chuck for unmounted stone or rubber lathe wheels, and shall be suitable for attachment to the left hand side of the machine.

3.4.4.7 *Chuck, spindle, left.*—Chuck, spindle, left, shall be a standard brass or bronze tapered spindle chuck for cloth, felt, and brush buffing wheels, and shall be suitable for attachment to the left hand side of machine.

3.5 Dimensions.—Unless otherwise specified, in the invitation for bids, the maximum dimensions of the machine, in inches, shall be as follows:

Over-all height	8 (without feet)
Over-all length of shaft	12⅝
Diameter of housing	6⁹⁄₁₆
Width of base	6½
Depth of base	7

3.6 Finish.—Unless otherwise specified, machine shall have a black enamel finish, except for unpainted parts which shall be chromium over nickel-plated. Chucks fabricated of carbon steel shall have a black oxide finish conforming to class 1, grade A of Military Specification MIL–F–13924. Chucks fabricated of brass or bronze shall have a lacquered satin finish.

3.7 Identification marking.—Each machine shall be provided with a readily visible metal nameplate permanently affixed thereto. The plate shall be marked with the name or registered trade-mark of the manufacturer, model designation, serial number, current, voltage, cycle, and unless otherwise specified, Stock No., OIN, and a caduceous centered between the letters "U" and "S". The caduceous shall be in accordance with Drawing 1056. (See figure 1). The markings on the plate shall be clear and permanent, and shall be accomplished by die stamping into the surface of the metal. Alternate methods of marking the nameplate (other than decalcomania) may be submitted to the contracting officer for approval.

3.8 Electrical parts.—All electrical parts and wiring shall conform to the requirements of the National Electrical Code, as amended, and the applicable standards of the Underwriters' Laboratories, Inc. (See 3.1).

3.9 Service data.—The contractor shall enclose with each unit two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number. In addition, the contractor shall mail six copies of the service data to the contracting officer for distribution.

3.10 Workmanship.—The machines, including adapters and chucks, shall be free from characteristics or defects which affect the appearance or which might affect the serviceability or render the machines unsuitable or inefficient for the intended purpose.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, samples shall be submitted in accordance with instructions furnished by the procuring activity (see 6.2). In addition, sampling procedures shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.2 Inspection.—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at the point of delivery, as specified (see 6.2). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

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4.3 Tests.—Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with this specification.

4.3.1 *Motor tests.*—Motors shall be tested in accordance with the applicable test requirements of Federal Specification CC–M–636. Motors shall also be tested for armature or rotor stall to determine compliance with the requirements of 3.4.3.12.

5. PREPARATION FOR DELIVERY

(For definitions and levels, see 6.1).

5.1 Packaging.

5.1.1 *Unit of issue.*—One (1) machine, as specified, constitutes one unit.

5.1.2 *Level A—military preparation.*—Exterior base metal surfaces subject to corrosion shall be treated with corrosion preventive conforming to grade 3 of Military Specification MIL–C–16173. All exterior finished surfaces shall be protected by wrapping with Kraft paper and cushioning with excelsior pads or flexible corrugated fiberboard or other suitable materials. Wrapping and cushioning material shall be secured by tying with cotton cord or tape, or by sealing with Kraft gummed paper tape. Work shall be accomplished in such manner as to insure against marring, scratching, or damage to the unit.

5.1.3 *Level B—limited military preparation.*—Preparation shall be as specified in 5.1.2.

5.1.4 *Level C—minimum military preparation.*—Exterior surfaces shall be protected by methods commonly used for the purpose which will afford the degree of protection required for shipment and use of the end product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A—military pack.*—Each unit shall be overpacked in an exterior container designed for type 3 load, and constructed in accordance with Military Specification JAN–P–106 or Federal Specification PPP–B–601, table II, using type III, class 1 plywood. Closure and strapping shall comply with appendix of applicable specification. Each box shall be lined with a waterproof barrier fabricated of material conforming to Military Specification JAN–P–125 and sealed with a suitable waterproof adhesive. The unit shall be adequately secured within the exterior container, by means of bolts, wood braces, blocks or other suitable devices, to insure against shifting and breakage, or marring of surfaces.

5.2.2 *Level B — limited military pack.*—Each unit shall be overpacked in an exterior container constructed in accordance with Federal Specification NN–B–621 or PPP–B–601, table I, using type III, class I plywood. The unit shall be adequately secured within the exterior container, by means of bolts, wood braces, blocks or other suitable devices, to insure against shifting and breakage, or marring of surfaces.

5.2.3 *Level C—minimum military pack.*—The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Level A—military marking.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color marking and symbol are not required on containers for Level A packs when addressed to domestic establishments for the Medical Services of the Armed Forces.

5.3.2 *Level B—limited military marking.*—Shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

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5.3.3 *Level C—minimum military marking.* —Shipping containers shall be marked with stock number, name of material, quantity, gross weight, name of consignee, name of consignor, and procurement instruction and office identifying numbers as shown on contract or purchase order.

5.3.4 *Name of items.*—The nomenclature used for identification purposes shall be as specified in 6.3.

6. NOTES

6.1 Federal Standard 102 should be referred to for definitions and application of the various levels of packaging protection for supplies and equipment.

6.2 Ordering data. — Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Sampling required (see 4.1).
(d) Inspection required (see 4.2).
(e) Place of final inspection (see 4.2).
(f) Level of protection required (see 5.1, 5.2, and 5.3).

6.3 This specification covers the following items appearing in the Armed Services Medical Stock List:

6520–538–9150 Grinding and Polishing Machine, Dental Laboratory, Bench Mounted, 110 volt, 60 cycle, AC.

6520–538–9850 Adapter, Chuck, Dental Grinding and Polishing Machine, Right, Automatic.

6520–539–0075 Chuck, Grinding and Polishing Machine, Dental Laboratory, Arbor Band, Right.

6520–539–0155 Chuck, Grinding and Polishing Machine, Dental Laboratory, Bur, Right.

6520–539–0250 Chuck, Grinding and Polishing Machine, Dental Laboratory, Left.

6520–539–0305 Chuck, Grinding and Polishing Machine, Dental Laboratory, Right.

6520–539–0350 Chuck, Grinding and Polishing Machine, Dental Laboratory, Stone, Left.

6520–539–0550 Chuck, Grinding and Polishing Machine Dental Laboratory, Stone, Right.

6520–539–1200 Spring, Dental Grinding and Polishing Machine Chuck Adapter.

6520–539–1225 Spring, Dental Grinding and Polishing Machine Bur Chuck.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.5 Transportation description.

6.5.1 *Transportation description applicable to this item is:*

Dental Machines
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.
Truckload minimum weight 20,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Army—M
Navy—Md
Air Force.

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FIGURE 1.—Insignia.

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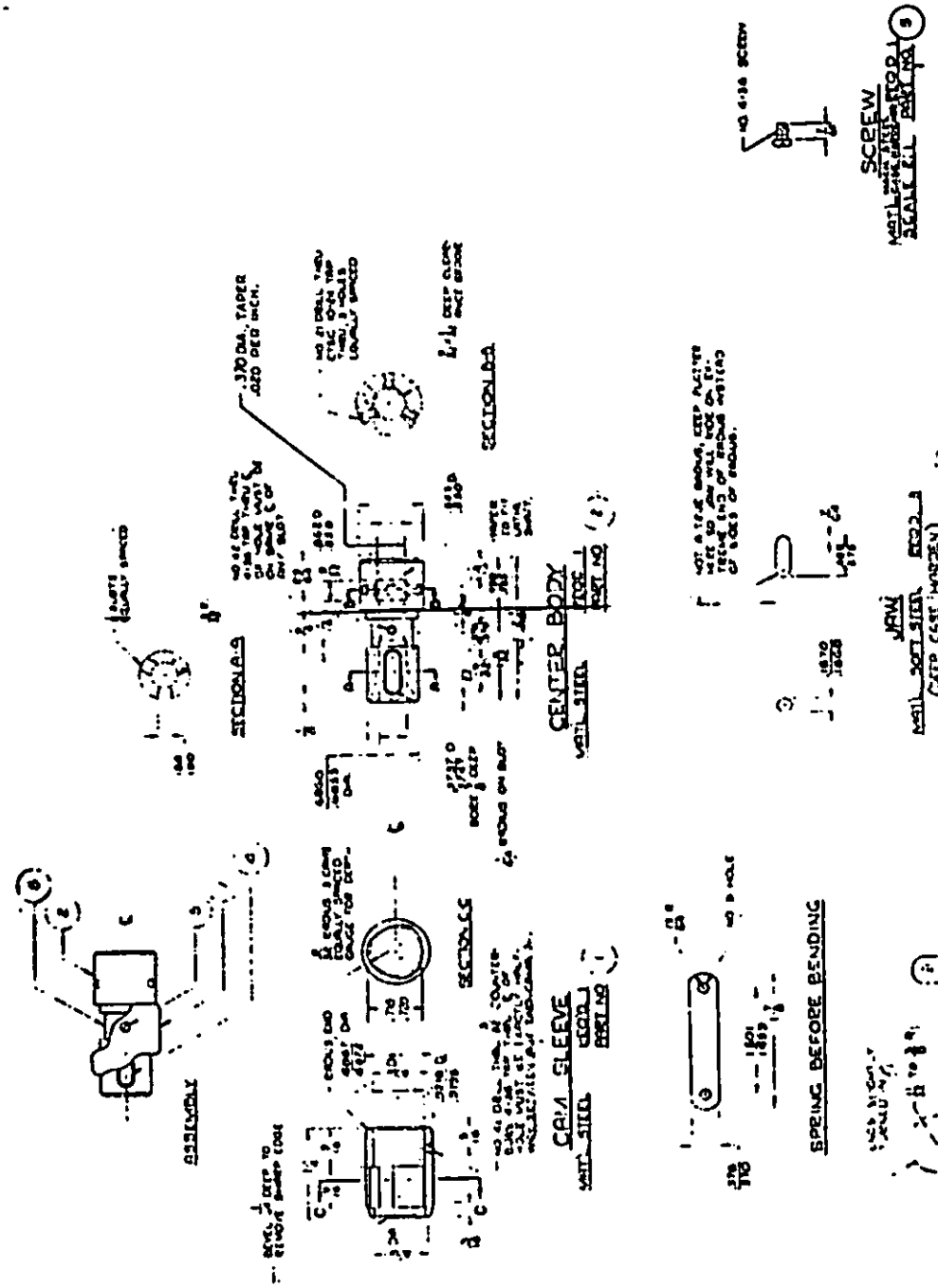
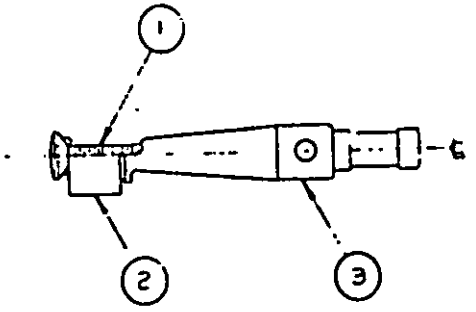
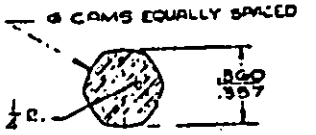


FIGURE 2.—Adapter, chuck, dental grinding and polishing machine, right, automatic.

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ASSEMBLY



SECTION A-A
SCALE 2:1

GENERAL NOTES

1- THIS ITEM FOR USE WITH FED. STK. No. 6520-558-9650.

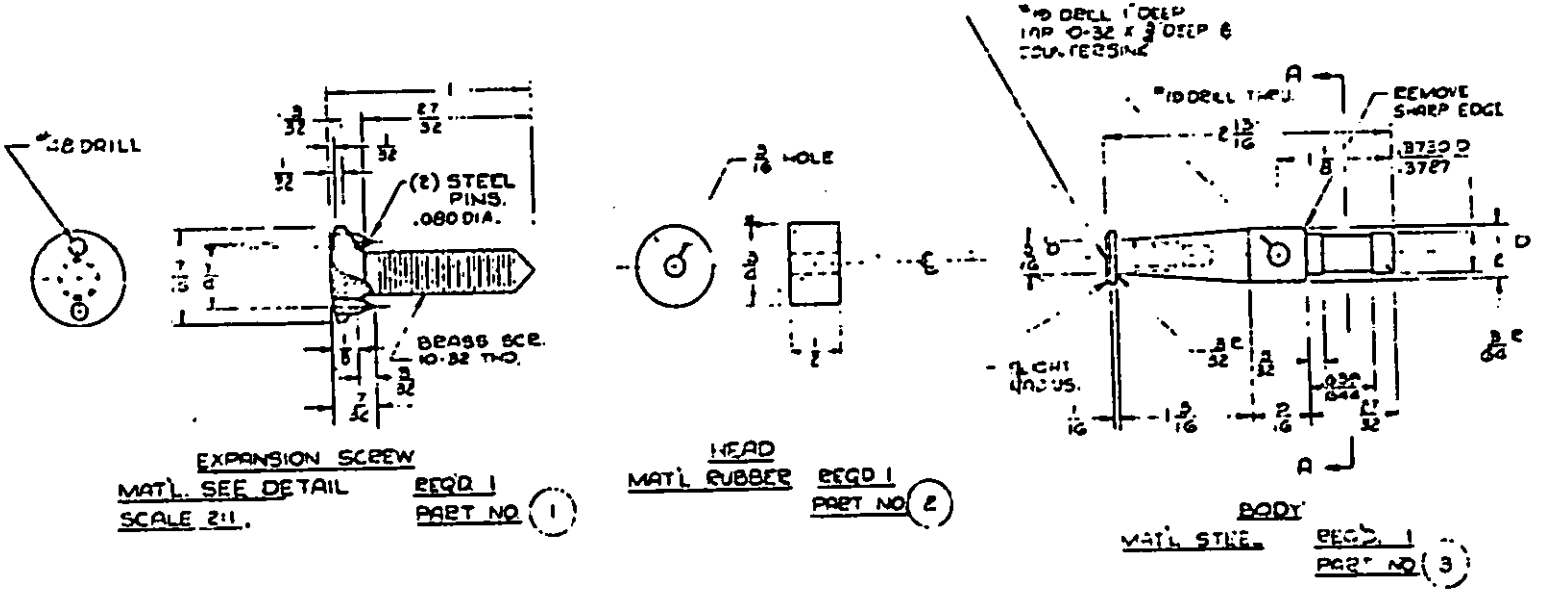


FIGURE 3.—Chuck, grinding and polishing machine, dental laboratory, arbor band, right.

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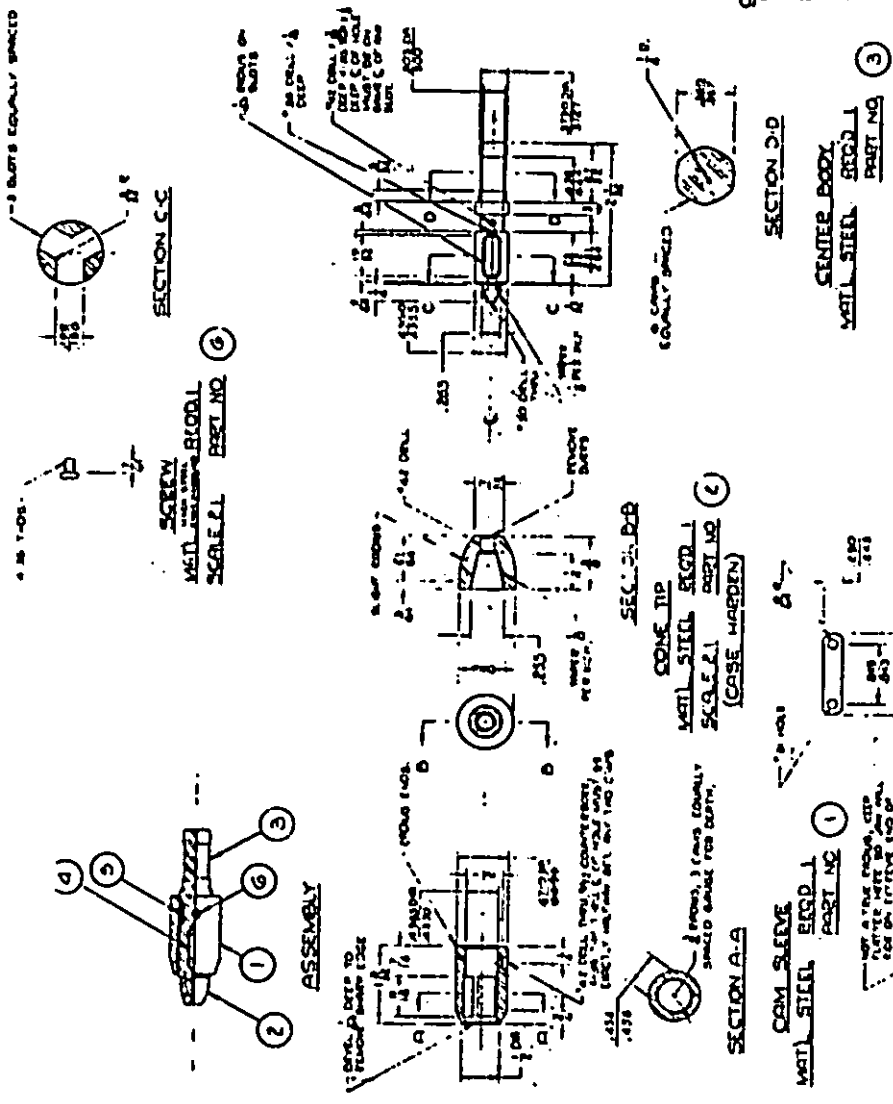


FIGURE 4.—Chuck, grinding and polishing machine, dental laboratory, bur, right.

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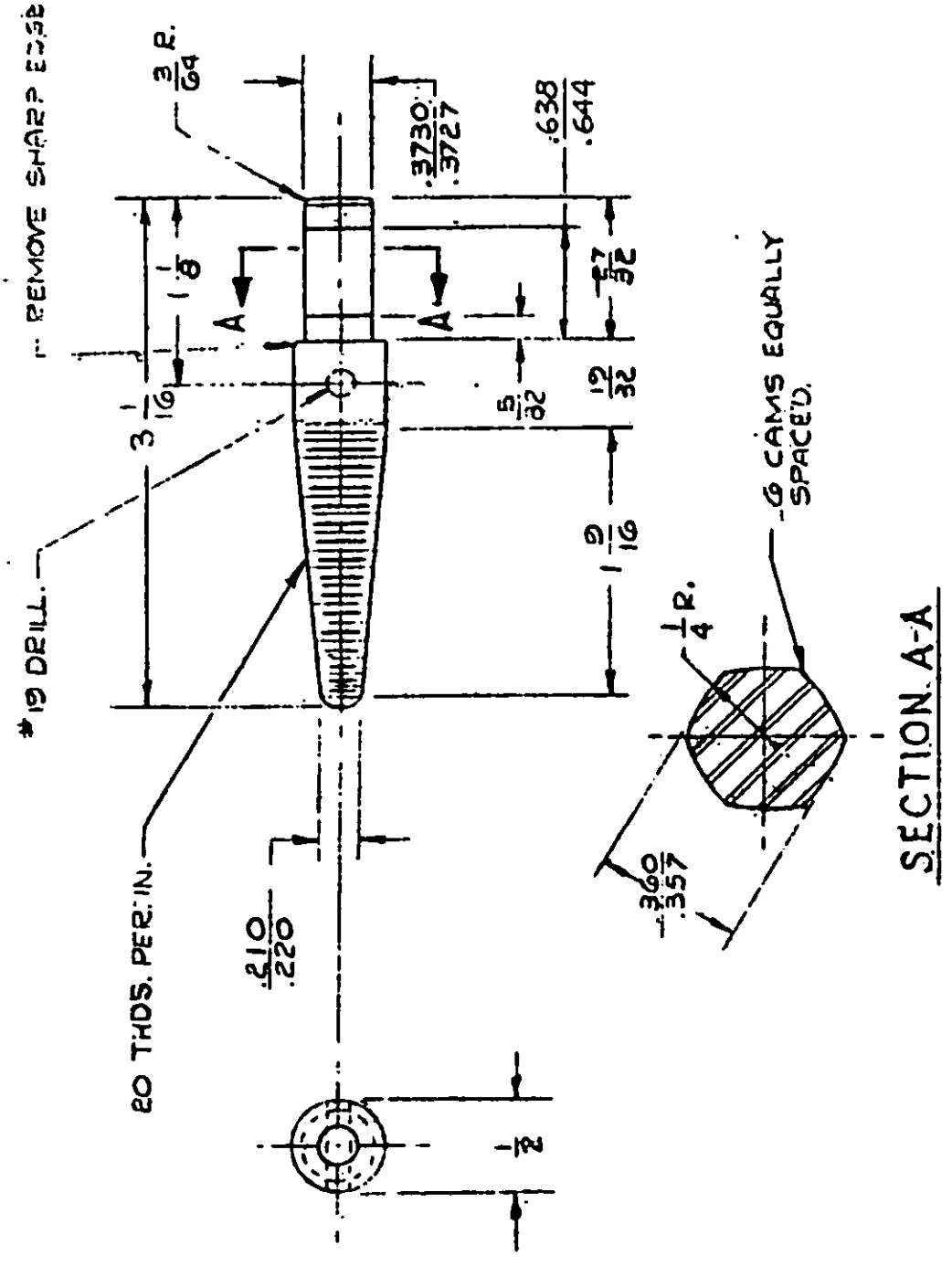


FIGURE 5.—Chuck, grinding and polishing machine, dental laboratory, right.

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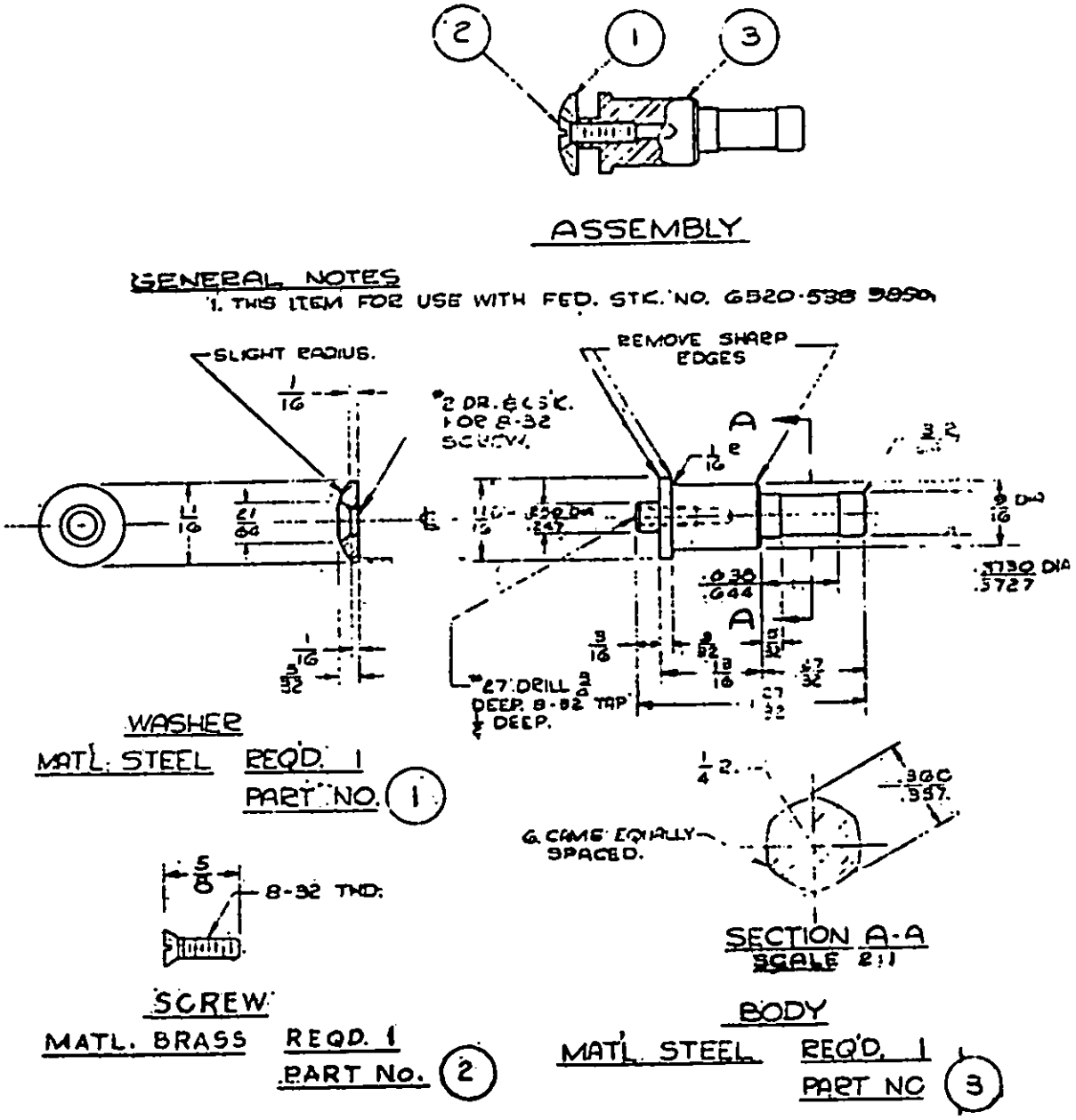


FIGURE 6.—Chuck, grinding and polishing machine, dental laboratory, stone, right.

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